



# Koneru Lakshmaiah Education Foundation

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## DEPARTMENT OF CSE

### PROJECT ABSTRACT SUBMISSION

### COURSE: ENGINEERING CAPSTONE PROJECT – 1(23IE4053R/23IE4053A) A.Y. 2026 - 2027

| TITLE OF THE PROJECT | AI-Driven Software Architecture and Codebase Intelligence |                    |
|----------------------|-----------------------------------------------------------|--------------------|
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## Abstract

The proposed system is an **AI-Driven Software Architecture and Codebase Intelligence** designed to help developers understand, evaluate, and maintain complex software projects. As software systems grow larger, manually identifying code quality issues, architectural dependencies, security vulnerabilities, and technical debt becomes increasingly challenging. The proposed solution combines artificial intelligence with automated software analysis to address these challenges efficiently.

The system analyzes a given codebase to identify **code smells, complexity, duplication, dependencies, maintainability issues, and security vulnerabilities** using techniques such as AST-based analysis, SonarQube, and Trivy. The results are presented through an interactive web interface, providing developers with a clear overview of the software's overall health and potential risks.

A **Generative AI component** enables natural-language interaction with the analyzed codebase. Developers can ask questions about potential problems, change impacts, security concerns, and possible improvements, receiving contextual explanations and recommendations.

The system also provides **architecture and dependency visualization** to help developers understand relationships between software components. By integrating AI, software engineering practices, and DevSecOps tools, the proposed system aims to reduce manual analysis effort, improve software quality and security, and support informed decision-making throughout the software development lifecycle.